

Grade 12 Prelim Paper 1
Memo

Question 1

1.1.1. $x^2 + 2x - 3 = 12$

$$x^2 + 2x - 15 = 0$$

$$(x+5)(x-3) = 0 \quad \checkmark A$$

$$x = -5 \quad \checkmark CA \quad \text{or} \quad x = 3 \quad \checkmark CA \quad (3)$$

1.1.2. $4x^2 + 3 = 9x$

$$4x^2 - 9x + 3 = 0 \quad \checkmark A$$

$$x = \frac{-(-9) \pm \sqrt{(-9)^2 - 4(4)(3)}}{2(4)} \quad \checkmark \text{Quad formula}$$

$$x \approx 1,84 \quad \checkmark CA \quad \text{or} \quad x \approx 0,41 \quad \checkmark CA \quad (4)$$

1.1.3. $2 - x - x^2 > 0$

$$x^2 + x - 2 < 0 \quad \checkmark$$

$$(x+2)(x-1) < 0$$

Critical values



CV: -2 or 1

$$\therefore -2 < x < 1 \quad \checkmark A$$

(4)

$$1.1.4. \quad x+1 = \sqrt{x+13}$$

$$(x+1)^2 = (\sqrt{x+13})^2 \quad \checkmark_m$$

$$x^2 + 2x + 1 = x + 13$$

$$x^2 + x - 12 = 0 \quad \checkmark_A$$

$$(x+4)(x-3) = 0 \quad \checkmark$$

$$x = -4$$

N/A $\checkmark_{\text{reject}}$

$\checkmark_{CA}$ both

$$\text{or } \underline{x = 3},$$

(5)

$$1.2. \quad x+y = 1$$

$$x^2 - y^2 - xy - y = 13 \dots (2)$$

$$x = 1 - y \dots (1) \quad \checkmark_m$$

Sub (1) into (2)

$$(1-y)^2 - y^2 - y(1-y) - y = 13 \quad \checkmark$$

$$1 - 2y + y^2 - y^2 - y + y^2 - y = 13$$

$$y^2 - 4y - 12 = 0$$

$$(y-6)(y+2) = 0 \quad \checkmark$$

$$y = 6$$

$\checkmark_m$

or

$$y = -2$$

$\checkmark_{CA}$ both x

$$\therefore x = -5$$

or

$$\therefore x = 3$$

(6)

$$1.3. \frac{27^a}{9^b} = 81$$

$$\frac{3^{3a}}{3^{2b}} = 81$$

✓ Prime base

$$3^{3a-2b} = 3^4 \quad \checkmark$$

$$\therefore 3a - 2b = 4 \quad \checkmark$$

(3)

Question 2

2.1. $-20 - 15 - 10 - \dots$

$$S_n = \frac{n}{2} [2a + (n-1)d]$$

$$225 = \frac{n}{2} [2(-20) + (n-1)(5)]$$

✓ A

✓ Correct subs into correct formula

$$225 = \frac{n}{2} [-40 + 5n - 5]$$

$$450 = n(5n - 45)$$

$$450 = 5n^2 - 45n$$

$$n^2 - 9n - 90 = 0$$

✓ Standard form

$$(n - 15)(n + 6) = 0$$

$$\underline{n = 15},$$

✓ CA both
or

$$n = -6$$

N/A ✓ reject

(5)

$$2.2. \quad -\frac{3}{2} + 3 - 6 + \dots - 96$$

$$T_n = ar^{n-1}$$

✓ Correct subs into correct formula

$$T_n = \left(-\frac{3}{2}\right) \left(-2\right)^{n-1} = -96 \quad \checkmark = -96$$

$$\left(-2\right)^{n-1} = 64$$

$$\left(-2\right)^{n-1} = \left(-2\right)^6$$

$$\therefore n-1 = 6$$

$$n = 7 \quad \checkmark$$

$$\therefore \sum_{k=1}^7 -\frac{3}{2} \left(-2\right)^{k-1} \quad \checkmark \text{CA}$$

(4)

$$2.3.1. \quad 1 + \frac{1}{2} + \frac{1}{4} + \dots$$

$$S_{\infty} = \frac{a}{1-r}$$

✓ correct subs into
correct formula

$$S_{\infty} = \frac{1}{1 - \frac{1}{2}} = 2 \quad \checkmark$$

(2)

$$2.3.2. \quad S_k = \frac{a(1-r^k)}{1-r}$$

$$= \frac{1(1 - (\frac{1}{2})^k)}{1 - \frac{1}{2}}$$

$$= 2(1 - (\frac{1}{2})^k)$$

$$= 2 - 2 \cdot (\frac{1}{2})^k \quad \checkmark S_k$$

$$S_{\infty} - S_k = 2 - (2 - 2 \cdot (\frac{1}{2})^k)$$

$$= 2 \cdot (\frac{1}{2})^k \quad \checkmark^{CA} \quad \text{Alt: } 2^{1-k}$$

$$2 \cdot (\frac{1}{2})^k = \frac{1}{1000} \quad \checkmark^m \quad \text{Alt: } 2^{1-k} = \frac{1}{1000}$$

$$(\frac{1}{2})^k = \frac{1}{2000}$$

$$1-k = \log_2 \frac{1}{1000}$$

$$1-k = -9,97$$

$$k = \log_{\frac{1}{2}} \frac{1}{2000} \quad \checkmark$$

$$k \approx 10,97$$

$$k \approx 10,97$$

$$\therefore \underline{k = 11} \rightarrow \checkmark^{CA}$$

$$\underline{k = 11} \rightarrow$$

(5)

Question 3

1st Difference: $3n-4 = -1; 2; 5; \dots$ ✓ pattern

2nd Difference = 3

$$2a = 3$$

✓³

$$3a + b = -1$$

$$a = \frac{3}{2} \quad \checkmark$$

$$3\left(\frac{3}{2}\right) + b = -1$$

$$\therefore b = -\frac{11}{2} \quad \checkmark$$

$$T_n = \frac{3}{2}n^2 - \frac{11}{2}n + c$$

$$\therefore \frac{3}{2}(31)^2 - \frac{11}{2}(31) + c = 1335 \quad \checkmark^m$$

$$\frac{2883}{2} - \frac{341}{2} + c = 1335$$

$$\therefore c = 64 \quad \checkmark_{CA}$$

$$\therefore T_n = \frac{3}{2}n^2 - \frac{11}{2}n + 64$$

Question 4

$$4.1. f(x) = \frac{1-x}{x+2}$$

$$= - \left(\frac{x-1}{x+2} \right)$$

$$= - \left(\frac{x+2-3}{x+2} \right) \checkmark$$

$$= - \left(1 - \frac{3}{x+2} \right) \checkmark$$

$$f(x) = -1 + \frac{3}{x+2}$$

(2)

4.2. On Answer sheet.

$$4.3. y \in \mathbb{R} \checkmark; y \neq -1 \checkmark$$

(2)

$$4.4. y+1 = -(x+2)$$

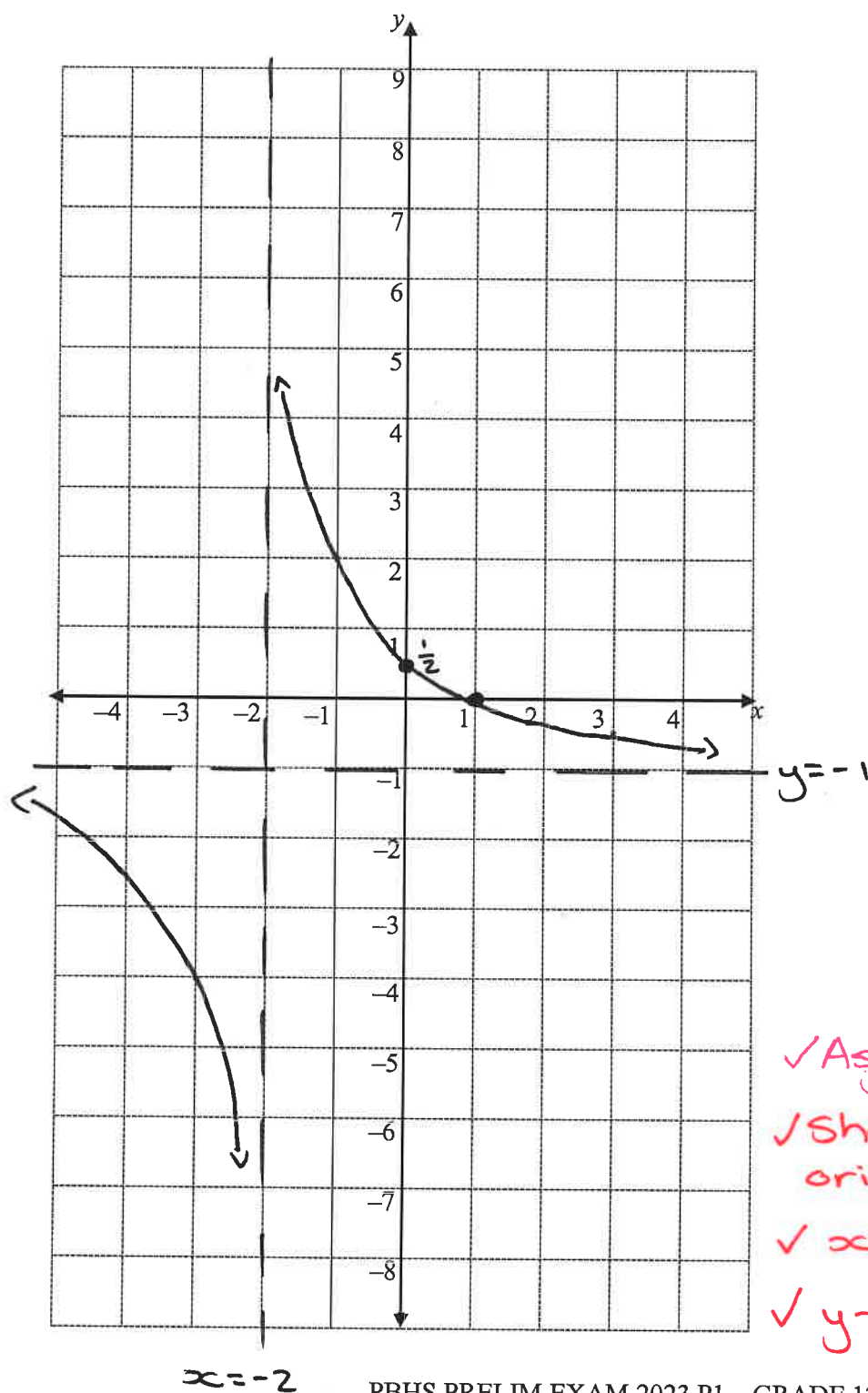
$$y = -x - 3 \checkmark$$

(2)

NAME: Memo

FORM CLASS: _____

QUESTION 4.2



- ✓ Asymptotes
- ✓ Shape + orientation
- ✓ x -int (1; 0)
- ✓ y -int (0; $\frac{1}{2}$)

$x = -2$

Question 5

$$5.1. x = \frac{-1}{2(2)} \quad \checkmark m$$

$$f\left(-\frac{1}{4}\right) = -\frac{49}{8}$$

$$x = -\frac{1}{4}$$

$$TP \left(-\frac{1}{4} ; -\frac{49}{8} \right)$$

Alt:

$$2 \left(x^2 + \frac{x}{2} - 3 \right)$$

$$= 2 \left[\left(x + \frac{1}{4} \right)^2 - \frac{49}{16} \right]$$

$$= 2 \left(x + \frac{1}{4} \right)^2 - \frac{49}{8}$$

$$\therefore TP \left(-\frac{1}{4} ; -\frac{49}{8} \right)$$

(3)

5.2. let $y=0$: $0 = 2x^2 + x - 6$ ✓_m

$$0 = (2x - 3)(x + 2)$$

$$x = \frac{3}{2} \quad \text{✓_{CA} or both} \quad x = -2$$

$$\therefore P(-2; 0)$$

$$Q\left(\frac{3}{2}; 0\right)$$

$$\therefore PQ = \frac{3}{2} - (-2) \quad \text{✓_m}$$

$$= \frac{7}{2} \approx 3,5 \text{ units} \quad \text{✓}$$

(4)

5.3. $m = 3$ ✓_A

(parallel lines)

$c = 6$ ✓_A

(shift ↑ 8)

(2)

5. 4. $2x^2 + x - 6 = 3x - 2$ ✓ $f(x) = g(x)$

$$2x^2 - 2x - 4 = 0$$

$$x^2 - x - 2 = 0$$

$$(x - 2)(x + 1) = 0$$

$$x = 2 \quad \text{✓_{CA} both} \quad \text{or } x = -1$$

$$\therefore x \leq -1 \text{ or } x \geq 2 \quad \text{✓_{CA} values}$$

✓ inequalities

(4)

$$5.5.1. \quad x \leq -2 \quad \checkmark \quad \text{or} \quad x \geq \frac{3}{2} \quad \checkmark \quad \text{Accept} \\ < > \text{ and} \\ \leq \geq$$

CA values from 5.2.

$$5.5.2. \quad h''(x) = f'(x) = 4x + 1$$

$$4x + 1 < 0 \quad \checkmark h'' < 0$$

$$4x < -1$$

$$x < -\frac{1}{4} \quad \checkmark \text{CA}$$

(2)

$$5.6. \quad f'(x) = 4x + 1$$

$$4x + 1 = 3 \quad \checkmark f'(x) = m = 3$$

$$4x = 2$$

$$x = \frac{1}{2} \quad \checkmark \text{CA}$$

$$f\left(\frac{1}{2}\right) = -5 \quad \checkmark \text{CA} \quad \text{Point of intersection: } \left(\frac{1}{2}; -5\right)$$

$$\therefore -5 = 3\left(\frac{1}{2}\right) + k \quad \checkmark m$$

$$\therefore k = -\frac{13}{2} \quad \checkmark \text{CA}$$

(5)

5.6 Alt:

$$2x^2 + x - 6 = 3x + k \quad \checkmark m$$

$$2x^2 - 2x - k - 6 = 0 \quad \checkmark A$$

One R Solution: $\Delta = 0$

$$b^2 - 4ac = 0 \quad \checkmark \Delta = 0$$

$$(-2)^2 - 4(2)(-k-6) = 0 \quad \checkmark$$

$$4 - 8(-k-6) = 0$$

$$4 + 8k + 48 = 0$$

$$8k = -52$$

$$k = -\frac{13}{2} \quad \checkmark$$

(5)

Question 6

6.1.1. $f^{-1}: x = 3y - 1$ ✓ $x \leftrightarrow y$

$$x + 1 = 3y \quad \checkmark$$

$$\therefore y = \frac{x + 1}{3} \quad \checkmark \quad \text{Alt: } y = \frac{x}{3} + \frac{1}{3} \quad (2)$$

6.1.2. $g^{-1}: x = 2^y$ ✓ $x \leftrightarrow y$

$$\therefore y = \log_2 x \quad \checkmark \quad (2)$$

6.2.1. $x < 1$ or $x > 3$ ✓ Inequalities ✓ Values (2)

6.2.2. $0 < x < 1$ ✓ ✓ (2)

6.2.3. $x = 2$ or $x = 8$ ✓ ✓ (2)

Question 7

$$7.1. 1 + i_{\text{eff}} = \left(1 + \frac{i_{\text{nom}}}{m}\right)^m \quad \checkmark$$

$$1 + 15\% = \left(1 + \frac{x}{2}\right)^2 \quad \checkmark$$

$$\frac{\sqrt{115}}{10} = 1 + \frac{x}{2}$$

$$\therefore x = 0,1447610...$$

$$\therefore x = 14,48\% \quad \checkmark$$

(3)

$$7.2. F = \frac{x[(1+i)^n - 1]}{i}$$

$$400\,000 = \frac{x \left[\left(1 + \frac{8\%}{4}\right)^{17} - 1 \right]}{\frac{8\%}{4}} \quad \checkmark n=17$$

✓ Correct Formula

✓ i; F

$$x = 19987,94 \quad \checkmark \text{CA}$$

(4)

$$7.3.1. P = \frac{x[1 - (1+i)^{-n}]}{i}$$

$$750\,000 = \frac{16\,000 \left[1 - \left(1 + \frac{12,5\%}{12} \right)^{-n} \right]}{\frac{12,5\%}{12}}$$

✓ Correct formula

✓ $i; x; P$

$$\frac{131}{256} = \left(\frac{97}{96} \right)^{-n}$$

$$-n = \log_{\frac{97}{96}} \left(\frac{131}{256} \right) \quad \checkmark m$$

$$n = 64,65 \quad \checkmark$$

∴ 65 payments ✓ CA

(5)

7.3.2) OB after 64 full payments:

$$OB = P(1+i)^m - \frac{x[(1+i)^m - 1]}{i} \quad \checkmark m$$

$$= 750\,000 \left(1 + \frac{12,5\%}{12}\right)^{64} - 16\,000 \left[\left(1 + \frac{12,5\%}{12}\right)^{64} - 1 \right] \frac{12,5\%}{12} \quad \checkmark m=64$$

$$OB = 10\,351,01461 \quad \checkmark$$

$$\text{Last Payment} = 10\,351,01461 \left(1 + \frac{12,5\%}{12}\right) \quad \checkmark$$

$$= 10\,458,84 \quad \checkmark$$

(5)

Question 8

$$8.1. f(x+h) = \frac{4}{x+h}$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \quad \checkmark$$

$$= \lim_{h \rightarrow 0} \frac{\frac{4}{x+h} - \frac{4}{x}}{h} \quad \checkmark$$

$$= \lim_{h \rightarrow 0} \frac{4x - 4(x+h)}{x(x+h)} \div h$$

$$= \lim_{h \rightarrow 0} \frac{-4h}{x^2 + xh} \times \frac{1}{h}$$

$$= \lim_{h \rightarrow 0} \frac{-4}{x^2 + xh} \quad \checkmark$$

$$= \frac{-4}{x^2 + x(0)}$$

$$= \frac{-4}{x^2} \quad \checkmark$$

(4)

$$8.2. \quad D_x \left[\frac{(\sqrt{x} - 1)^2}{\sqrt{x}} \right]$$

$$D_x \left[\frac{x - 2\sqrt{x} + 1}{\sqrt{x}} \right] \quad \checkmark \text{Product}$$

$$D_x \left[\sqrt{x} - 2 + \frac{1}{\sqrt{x}} \right]$$

$$D_x \left[x^{\frac{1}{2}} - 2 + x^{-\frac{1}{2}} \right] \quad \checkmark \text{Roots} \rightarrow \text{Exponents}$$

$$= \frac{1}{2} x^{-\frac{1}{2}} - \frac{1}{2} x^{-\frac{3}{2}}$$

(4)

Question 9

9.1. $f(x) = a(x - x_1)^2(x - x_2)$

$f(x) = -1(x - 2)^2(x + 3)$ ✓ x -ints

$$= -1(x^2 - 4x + 4)(x + 3)$$

$$= -(x^3 - 4x^2 + 4x + 3x^2 - 12x + 12)$$

$$= -(x^3 - x^2 - 8x + 12)$$

$$= -x^3 + x^2 + 8x - 12$$

(3)

$\therefore a = 1 \quad b = 8 \quad c = -12$

9.2. $f'(x) = -3x^2 + 2x + 8$ ✓ f'

$$-3x^2 + 2x + 8 = 0 \quad \checkmark f' = 0$$

$$3x^2 - 2x - 8 = 0$$

$$(3x + 4)(x - 2) = 0$$

$$x = -\frac{4}{3} \quad \checkmark \text{CA both} \quad \text{or} \quad x = 2$$

$$\therefore f\left(-\frac{4}{3}\right) = -\frac{500}{27}$$

$$\therefore A\left(-\frac{4}{3}; -\frac{500}{27}\right)$$

(4)

$$9.3. f''(x) = -6x + 2 \quad \checkmark f''$$

$$-6x + 2 = 0 \quad \checkmark f'' = 0$$

$$-6x = -2$$

$$x = \frac{1}{3} \quad \checkmark \text{CA}$$

(3)

$$9.4. -\frac{4}{3} < x < 2 \quad \checkmark \text{CA from 9.2} \quad \text{or } x < -3 \quad \checkmark$$

(2)

Question 10

10.1. $x^2 + 4xh = 1200$ ✓ SA = 1200

$$4xh = 1200 - x^2$$

$$h = \frac{1200 - x^2}{4x}$$
 ✓ $h = \dots$

$$V = x^2 h$$
 ✓ Volume

$$V = x^2 \left(\frac{1200 - x^2}{4x} \right)$$
 ✓ sub h in

$$V = \frac{1200x^2}{4x} - \frac{x^4}{4x}$$

$$V = 300x - \frac{x^3}{4}$$
 (4)

10.2. $V' = 300 - \frac{3}{4}x^2 = 0$ ✓ $V' = 0$

$$\frac{3}{4}x^2 = 300$$

$$3x^2 = 1200$$

$$x^2 = 400$$

$$x = 20\text{cm}$$
 ✓ CA

∴ Max Volume:

$$\therefore V(20) = 300(20) - \frac{(20)^3}{4} = 4000\text{cm}^3$$
 (5)

Question 11

$$11.1. \quad 3y + 4y + 5 + 6 + 3 + 14 + 17 + 45 = 160 \checkmark$$

$$7y = 70$$

$$y = 10 \checkmark$$

(2)

$$11.2.1. \quad P(R) = \frac{44 \checkmark}{160 \checkmark} = \frac{11}{40}$$

(2)

$$11.2.2. \quad P(\text{none}) = \frac{45 \checkmark}{160 \checkmark} = \frac{9}{32}$$

(2)

$$11.2.3. \quad P(\text{at least } 2) = \frac{28 \checkmark \checkmark}{160 \checkmark} = \frac{7}{40}$$

(3)

Question 12

12.1. $\frac{9!}{2! \times 2!}$ ✓A

= 90720 ✓CA (2)

12.2. Treat as identical:

$$\frac{\frac{7!}{2!}}{\frac{9!}{2! \times 2!}} \quad \checkmark \quad \checkmark \text{CA from 12.1}$$

= $\frac{1}{36}$ ✓ (3)

Alt: treat as not identical

$$\frac{1 \times 7! \times 2}{9!} \quad \checkmark$$

= $\frac{1}{36}$ ✓